Algebra: Partial Fractions & Division

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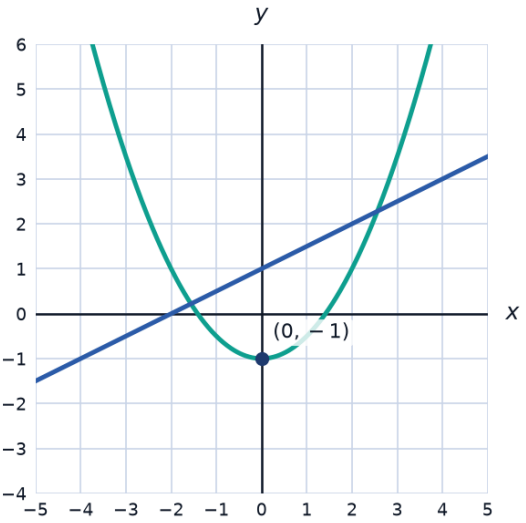
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Polynomial Division & Improper Fractions

The rational function f is defined for all real $xotin\{-1, 2\}$ by $f(x) = \frac{x^3 + kx + 7}{(x+1)(x-2)^2}$ where k is a constant. The partial fraction decomposition of $f(x)$ takes the form $A + \frac{B}{x+1} + \frac{C}{x-2} + \frac{D}{(x-2)^2}$ where $A, B, C,$ and D are real constants. Given that $C = 2$, find the value of k.

2 [Foundational] • Target: Partial Fractions with Linear & Repeated Factors

Express $\frac{-11x - 44}{(x+3)(x-8)}$ in the form $\frac{A}{(x+3)} + \frac{B}{(x-8)}$. Find B.



3 [Foundational] • Target: Irreducible Quadratic Factors

Express $\frac{-14x - 30}{(x-3)(x+6)}$ in the form $\frac{A}{(x-3)} + \frac{B}{(x+6)}$. Find B.

4 [Foundational] • Target: Binomial Series with Partial Fractions

Express $\frac{-14x - 22}{(x+1)(x+3)}$ in the form $\frac{A}{(x+1)} + \frac{B}{(x+3)}$. Find B.

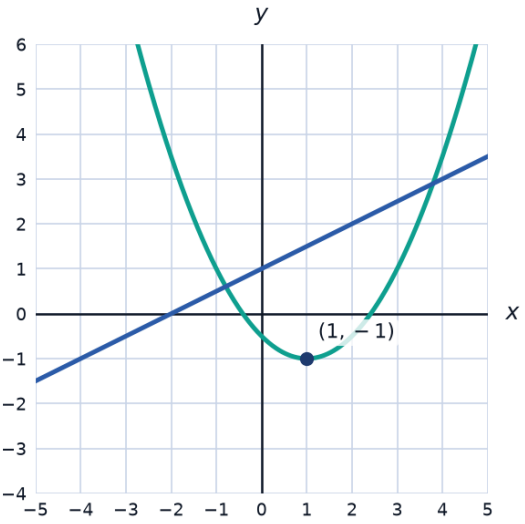
5 [Foundational] • Target: Polynomial Division & Improper Fractions

Express $\frac{-3x + 8}{(x-2)(x-4)}$ in the form $\frac{A}{(x-2)} + \frac{B}{(x-4)}$. Find B.

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6 [Foundational] • Target: Partial Fractions with Linear & Repeated Factors

Express $\frac{-2x-11}{(x+5)(x+4)}$ in the form $\frac{A}{(x+5)}+\frac{B}{(x+4)}$. Find A.



7 [Foundational] • Target: Irreducible Quadratic Factors

Express $\frac{-9x-11}{(x+1)(x+3)}$ in the form $\frac{A}{(x+1)}+\frac{B}{(x+3)}$. Find A.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Binomial Series with Partial Fractions

Express $\frac{3x-99}{(x-9)(x-1)}$ in the form $\frac{A}{(x-9)}+\frac{B}{(x-1)}$. Find B.

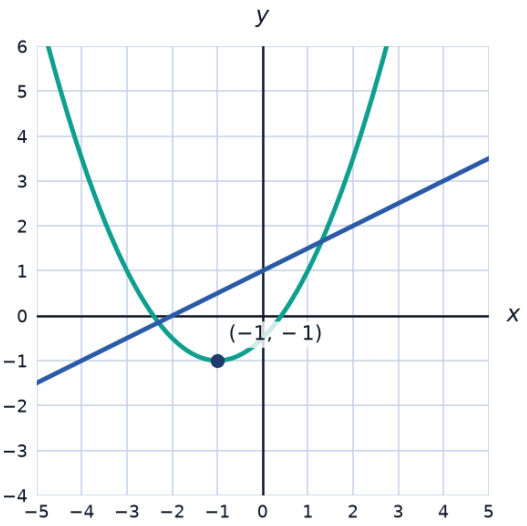
9 [Intermediate] • Target: Polynomial Division & Improper Fractions

Express $\frac{-7x+67}{(x+2)(x-7)}$ in the form $\frac{A}{(x+2)}+\frac{B}{(x-7)}$. Find A.

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10 [Intermediate] • Target: Partial Fractions with Linear & Repeated Factors

Express $\frac{2x-10}{(x-3)(x-1)}$ in the form $\frac{A}{(x-3)}+\frac{B}{(x-1)}$. Find A.



11 [Intermediate] • Target: Irreducible Quadratic Factors

Express $\frac{4x-36}{(x-3)(x+3)}$ in the form $\frac{A}{(x-3)}+\frac{B}{(x+3)}$. Find B.

12 [Intermediate] • Target: Binomial Series with Partial Fractions

Express $\frac{16x-94}{(x-9)(x+1)}$ in the form $\frac{A}{(x-9)}+\frac{B}{(x+1)}$. Find A.

13 [Intermediate] • Target: Polynomial Division & Improper Fractions

Express $\frac{2x-33}{(x-4)(x+1)}$ in the form $\frac{A}{(x-4)}+\frac{B}{(x+1)}$. Find A.

14 [Intermediate] • Target: Partial Fractions with Linear & Repeated Factors

Express $\frac{11x-6}{(x-6)(x+4)}$ in the form $\frac{A}{(x-6)}+\frac{B}{(x+4)}$. Find A.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

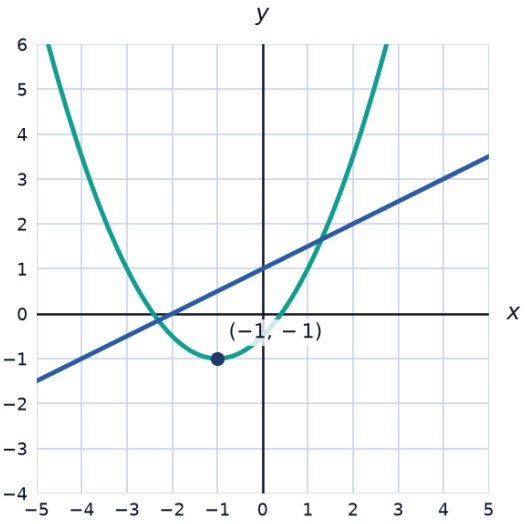
15 [Challenging] • Target: Irreducible Quadratic Factors • Cross-Lesson: 9.01 Complex Numbers

Express $\frac{2x-8}{(x-6)(x-2)}$ in the form $\frac{A}{(x-6)}+\frac{B}{(x-2)}$. Find B.

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16 [Challenging] • Target: Binomial Series with Partial Fractions • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Express $\frac{13x+34}{(x+2)(x+4)}$ in the form $\frac{A}{(x+2)}+\frac{B}{(x+4)}$. Find B.



17 [Challenging] • Target: Polynomial Division & Improper Fractions • Cross-Lesson: 10.01 Numerical Methods

Express $\frac{5x+24}{(x+4)(x+5)}$ in the form $\frac{A}{(x+4)}+\frac{B}{(x+5)}$. Find B.

18 [Challenging] • Target: Partial Fractions with Linear & Repeated Factors • Cross-Lesson: 11.01 Differential Equations

Express $\frac{3x+10}{(x+4)(x+2)}$ in the form $\frac{A}{(x+4)}+\frac{B}{(x+2)}$. Find B.

19 [Challenging] • Target: Irreducible Quadratic Factors • Cross-Lesson: 12.01 Poisson Distribution

Express $\frac{7x+15}{(x+2)(x+3)}$ in the form $\frac{A}{(x+2)}+\frac{B}{(x+3)}$. Find A.

20 [Challenging] • Target: Binomial Series with Partial Fractions • Cross-Lesson: 13.01 Continuous Random Variables

Express $\frac{9x+39}{(x+3)(x+5)}$ in the form $\frac{A}{(x+3)}+\frac{B}{(x+5)}$. Find B.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Polynomial Division & Improper Fractions

Divide $2x^3-3x^2+4x-5$ by $(x-2)$ to find the quotient and remainder.

a. Quotient $=2x^2+x+6$, Remainder $=7$

b. Quotient $=2x^2-7x+18$, Remainder $=-41$

c. Quotient $=2x^2+x+6$, Remainder $=-7$

d. Quotient $=2x^2+7x+18$, Remainder $=31$

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22 [Foundational] • Target: Polynomial Division & Improper Fractions

Express the improper rational function $\frac{x^2+5x+7}{x+2}$ in the form $Ax+B+\frac{C}{x+2}$.

a. $x+3+\frac{1}{x+2}$

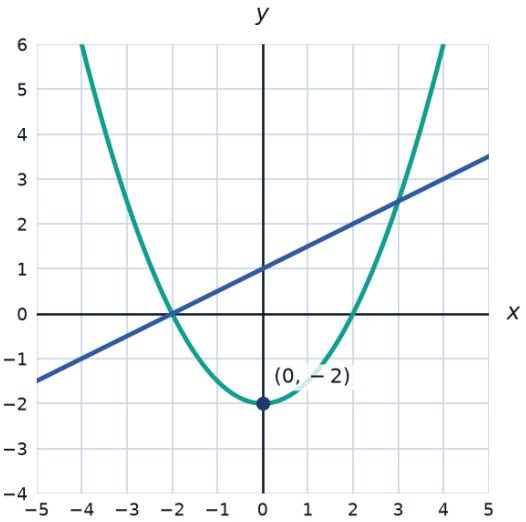
b. $x+3+\frac{7}{x+2}$

c. $x+7-\frac{7}{x+2}$

d. $x+2+\frac{3}{x+2}$

23 [Foundational] • Target: Polynomial Division & Improper Fractions

When $P(x)=x^4+ax^2+bx-4$ is divided by $(x-1)$, the remainder is 2; when divided by $(x+2)$, the remainder is -8. Find a and b.



a. $a=1, b=4$

b. $a=2, b=3$

c. $a=-1, b=6$

d. $a=3, b=2$

24 [Foundational] • Target: Partial Fractions with Linear & Repeated Factors

Express $\frac{7x-1}{(x-1)(x+2)}$ in partial fractions.

a. $\frac{2}{x-1}+\frac{5}{x+2}$

b. $\frac{5}{x-1}+\frac{2}{x+2}$

c. $\frac{6}{x-1}+\frac{1}{x+2}$

d. $\frac{7}{x-1}-\frac{1}{x+2}$

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25 [Foundational] • Target: Partial Fractions with Linear & Repeated Factors

What is the correct partial fraction decomposition template for $\frac{3x^2+1}{(x-2)^2(x+3)}$?

a. $\frac{A}{x-2}+\frac{B}{(x-2)^2}+\frac{C}{x+3}$

b. $\frac{A}{(x-2)^2}+\frac{B}{x+3}$

c. $\frac{Ax+B}{(x-2)^2}+\frac{C}{x+3}$

d. $\frac{A}{x-2}+\frac{B}{x+3}$

26 [Foundational] • Target: Partial Fractions with Linear & Repeated Factors

In the decomposition $\frac{4x}{(x-1)^2}=\frac{A}{x-1}+\frac{B}{(x-1)^2}$, find A and B.

a. $A=4, B=4$

b. $A=4, B=0$

c. $A=0, B=4$

d. $A=2, B=4$

27 [Foundational] • Target: Irreducible Quadratic Factors

What is the correct partial fraction template for $\frac{2x+5}{(x-1)(x^2+4)}$?

a. $\frac{A}{x-1}+\frac{Bx+C}{x^2+4}$

b. $\frac{A}{x-1}+\frac{B}{x^2+4}$

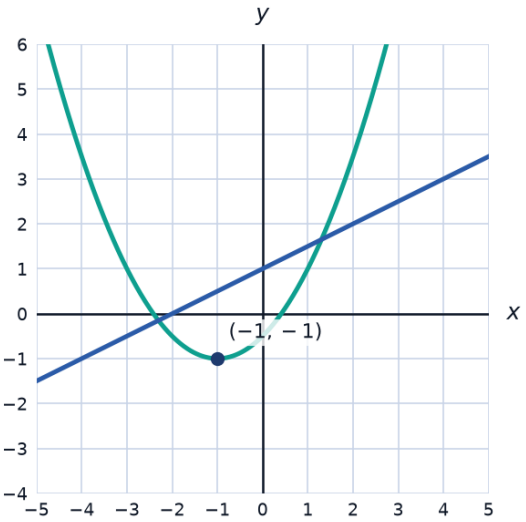
c. $\frac{Ax+B}{x-1}+\frac{C}{x^2+4}$

d. $\frac{A}{x-1}+\frac{B}{x+2}+\frac{C}{x-2}$

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28 [Foundational] • Target: Irreducible Quadratic Factors

In $\frac{7x^2+5x+13}{(x-1)(x^2+4)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+4}$, find the value of A.



a. 5

b. 7

c. 2

d. 3

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Irreducible Quadratic Factors

For the decomposition in the previous question ($A = 5$), find B and C.

a. $B = 2, C = 7$

b. $B = 2, C = -7$

c. $B = 7, C = 2$

d. $B = -2, C = 7$

30 [Intermediate] • Target: Binomial Series with Partial Fractions

To expand $f(x) = \frac{7x-1}{(x-1)(x+2)}$ in ascending powers of x for small $|x|$, which partial fraction form should be expanded?

a. $-\frac{2}{1-x} + \frac{5/2}{1+x/2}$

b. $\frac{2}{x-1} + \frac{5}{x+2}$

c. $\frac{2}{1-x} + \frac{5}{1+x/2}$

d. $-\frac{2}{1-x} + \frac{5}{1+x}$

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31 **[Intermediate]** • **Target: Binomial Series with Partial Fractions**

Find the constant term in the expansion of $\frac{7x-1}{(x-1)(x+2)}$ in ascending powers of x.

a. $\frac{1}{2}$

b. $-\frac{1}{2}$

c. -2

d. 5

32 **[Intermediate]** • **Target: Binomial Series with Partial Fractions**

State the radius of convergence for the expansion of $\frac{7x-1}{(x-1)(x+2)}$ in ascending powers of x.

a. $|x| < 1$

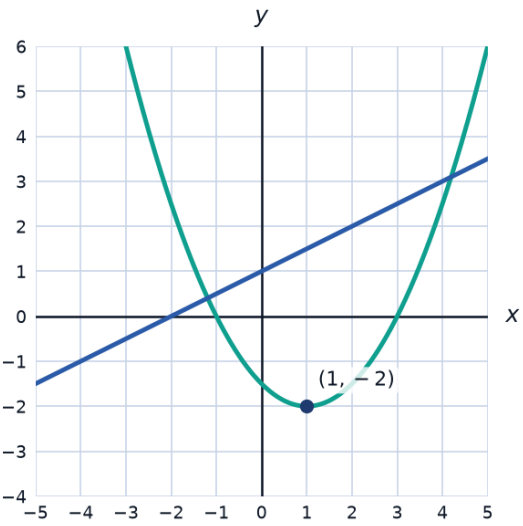
b. $|x| < 2$

c. $|x| < 1/2$

d. $|x| < 7$

33 **[Intermediate]** • **Target: Polynomial Division & Improper Fractions**

Which of the following is the correct expression of $\frac{5x+1}{(x-1)(x+2)}$ in partial fractions?



a. $\frac{2}{x-1}+\frac{3}{x+2}$

b. $\frac{3}{x-1}+\frac{2}{x+2}$

c. $\frac{2}{x-1}-\frac{3}{x+2}$

d. $\frac{1}{x-1}+\frac{4}{x+2}$

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34 **[Intermediate]** • **Target: Partial Fractions with Linear & Repeated Factors**

The rational function $f(x)$ is defined by $f(x) = \frac{5x+7}{(x-1)(x+3)}, \quad x \neq 1, \ x \neq -3$. Which of the following gives $f(x)$ expressed in partial fractions?

a. $\frac{3}{x-1} + \frac{2}{x+3}$

b. $\frac{2}{x-1} + \frac{3}{x+3}$

c. $\frac{3}{x-1} - \frac{2}{x+3}$

d. $\frac{4}{x-1} + \frac{1}{x+3}$

35 **[Intermediate]** • **Target: Irreducible Quadratic Factors**

Express $\frac{6x+27}{(x+2)(x+5)}$ in the form $\frac{A}{(x+2)} + \frac{B}{(x+5)}$. Find B.

a. 2

b. 1

c. -1

d. 5

36 **[Intermediate]** • **Target: Binomial Series with Partial Fractions**

Express $\frac{5x+19}{(x+5)(x+2)}$ in the form $\frac{A}{(x+5)} + \frac{B}{(x+2)}$. Find A.

a. 6

b. -2

c. 3

d. 2

37 **[Intermediate]** • **Target: Polynomial Division & Improper Fractions**

Express $\frac{5x+11}{(x+3)(x+2)}$ in the form $\frac{A}{(x+3)} + \frac{B}{(x+2)}$. Find B.

a. 2

b. 1

c. 4

d. -1

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Partial Fractions with Linear & Repeated Factors • Cross-Lesson: 11.01 Differential Equations

Express $\frac{2x-8}{(x+3)(x+1)}$ in the form $\frac{A}{(x+3)}+\frac{B}{(x+1)}$. Find A.

a. -5

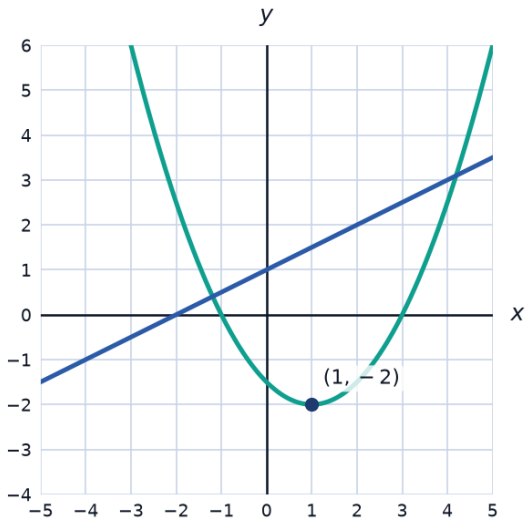
b. -7

c. 8

d. 7

39 [Challenging] • Target: Irreducible Quadratic Factors • Cross-Lesson: 12.01 Poisson Distribution

Express $\frac{17x+7}{(x+1)(x-1)}$ in the form $\frac{A}{(x+1)}+\frac{B}{(x-1)}$. Find B.



a. 5

b. 12

c. -12

d. 13

40 [Challenging] • Target: Binomial Series with Partial Fractions • Cross-Lesson: 13.01 Continuous Random Variables

Express $\frac{8x+23}{(x+1)(x+4)}$ in the form $\frac{A}{(x+1)}+\frac{B}{(x+4)}$. Find B.

a. -3

b. 3

c. 5

d. 4

41 [Challenging] • Target: Polynomial Division & Improper Fractions • Cross-Lesson: 14.01 Sampling & Estimation

Express $\frac{7x+22}{(x+1)(x+4)}$ in the form $\frac{A}{(x+1)}+\frac{B}{(x+4)}$. Find B.

a. 3

b. 5

c. -2

d. 2

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42 [Challenging] • Target: Partial Fractions with Linear & Repeated Factors • Cross-Lesson: 15.01 Hypothesis Testing

Express $\frac{8x+10}{(x-1)(x+2)}$ in the form $\frac{A}{(x-1)}+\frac{B}{(x+2)}$. Find B.

a. 3

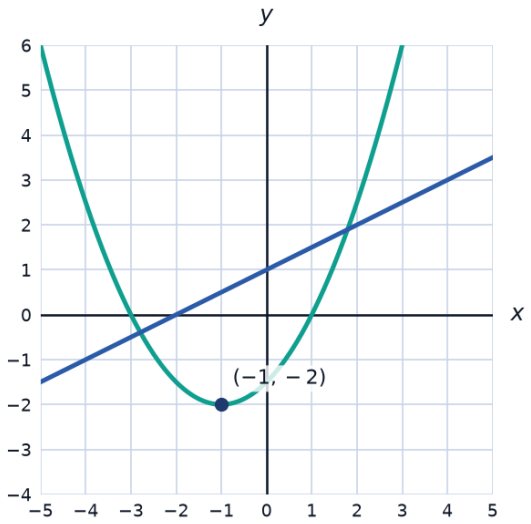
b. 6

c. 2

d. -2

43 [Challenging] • Target: Irreducible Quadratic Factors • Cross-Lesson: 2.01 Exponentials & Logarithms

Express $\frac{8x+32}{(x+5)(x+1)}$ in the form $\frac{A}{(x+5)}+\frac{B}{(x+1)}$. Find A.



a. -2

b. 6

c. 2

d. 3

44 [Challenging] • Target: Binomial Series with Partial Fractions • Cross-Lesson: 3.01 Advanced Trigonometry

Express $\frac{2x^2-94x-812}{(x+7)^2(x-7)}$ in the form $\frac{A}{(x+7)}+\frac{B}{(x+7)^2}+\frac{C}{(x-7)}$. Find B.

a. 9

b. -7

c. 4

d. 5

45 [Challenging] • Target: Polynomial Division & Improper Fractions • Cross-Lesson: 4.01 Differentiation

Express $\frac{7x+9}{(x+1)(x+2)}$ in the form $\frac{A}{(x+1)}+\frac{B}{(x+2)}$. Find A.

a. 3

b. 5

c. -2

d. 2

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